Arista Access Node BAS-ACC-78 QoS Congestion Event Resolution SOP

Node: BAS-ACC-78 | **Vendor:** Arista | **Severity:** Warning | **Alarm Code:** ALARM_1915

Description

This Standard Operating Procedure (SOP) outlines the steps to diagnose and resolve a QoS congestion event on Arista Access Node BAS-ACC-78, which is experiencing Microbursts, interface tail drops, and QoS queue starvation. The root cause of this failure is attributed to the misconfiguration of Quality of Service (QoS) policies, leading to an imbalance in bandwidth allocation and subsequent congestion on the access interface. The QoS policies, which are defined using the Modular QoS CLI (MQC), are applied to the ingress and egress traffic streams on the access interface. However, the configuration may not account for varying traffic patterns, resulting in the interface experiencing congestion and ultimately leading to the QoS queue starvation. To mitigate this, it is essential to monitor the QoS policies and adjust the bandwidth allocation as needed to prevent congestion and ensure optimal network performance.

Pre-requisite Checks

- check: Verify the site location and node ID at Basavanagudi POP

- check: Confirm the redundant link status is up and operational

- check: Ensure remote hands are dispatched to the site for on-site assistance (if required)

Potential Root Causes

Cause	Probability
Misconfigured QoS policies	High
Inadequate bandwidth allocation	Medium
Interface congestion due to Microbursts	Low

Diagnostic Commands & Expected Outputs

```
Command: show qos policy-map interface Ethernet1/1, Expected Output: Policy-map applied to interface Ethernet1/1, Description: This command displays the QoS policy-map applied to the access interface, allowing the technician to verify if the correct policy is applied and if the configuration is adequate.

Command: show interface statistics interface Ethernet1/1, Expected Output: Interface statistics, including packet drops and errors, Description: This command displays the interface statistics, including packet drops and errors, which can help identify if the interface is experiencing congestion.
```

```
Command: show mpls traffic-engineering topology, Expected Output: MPLS traffic engineering topology information, Description: This command displays the MPLS traffic engineering topology information, which can help identify if LDP or RSVP-TE is configured on the interface.
```

Resolution Steps

Step 1: Step: Verify QoS policy configuration, Command: show qos policy-map interface Ethernet1/1, Description: Verify the QoS policy-map applied to the access interface and ensure it is adequate to handle the varying traffic patterns.

Step 2: Step: Adjust bandwidth allocation, Command: policy-map Ethernet1/1, Description: Adjust the bandwidth allocation on the QoS policy-map to ensure sufficient bandwidth is allocated to prevent congestion on the interface.

Step 3: Step: Verify interface configuration, Command: show interface statistics interface Ethernet1/1, Description: Verify the interface configuration and ensure it is not experiencing congestion or packet drops.

Post-Check Verification

```
Command: show qos policy-map interface Ethernet1/1, Expected Output: Policy-map applied to interface Ethernet1/1, Description: Verify the QoS policy-map applied to the access interface and ensure it is adequate to handle the varying traffic patterns.

Command: show interface statistics interface Ethernet1/1, Expected Output: Interface statistics, including packet drops and errors, Description: Verify the interface configuration and ensure it is not experiencing congestion or packet drops.
```

Roll-back Procedures

- Step: Restore previous QoS policy configuration, Command: rollback to previous policy-map configuration, Description: Restore the previous QoS policy configuration to the access interface to prevent further congestion.

- Step: Verify interface configuration, Command: show interface statistics interface Ethernet1/1, Description: Verify the interface configuration and ensure it is not experiencing congestion or packet drops.